

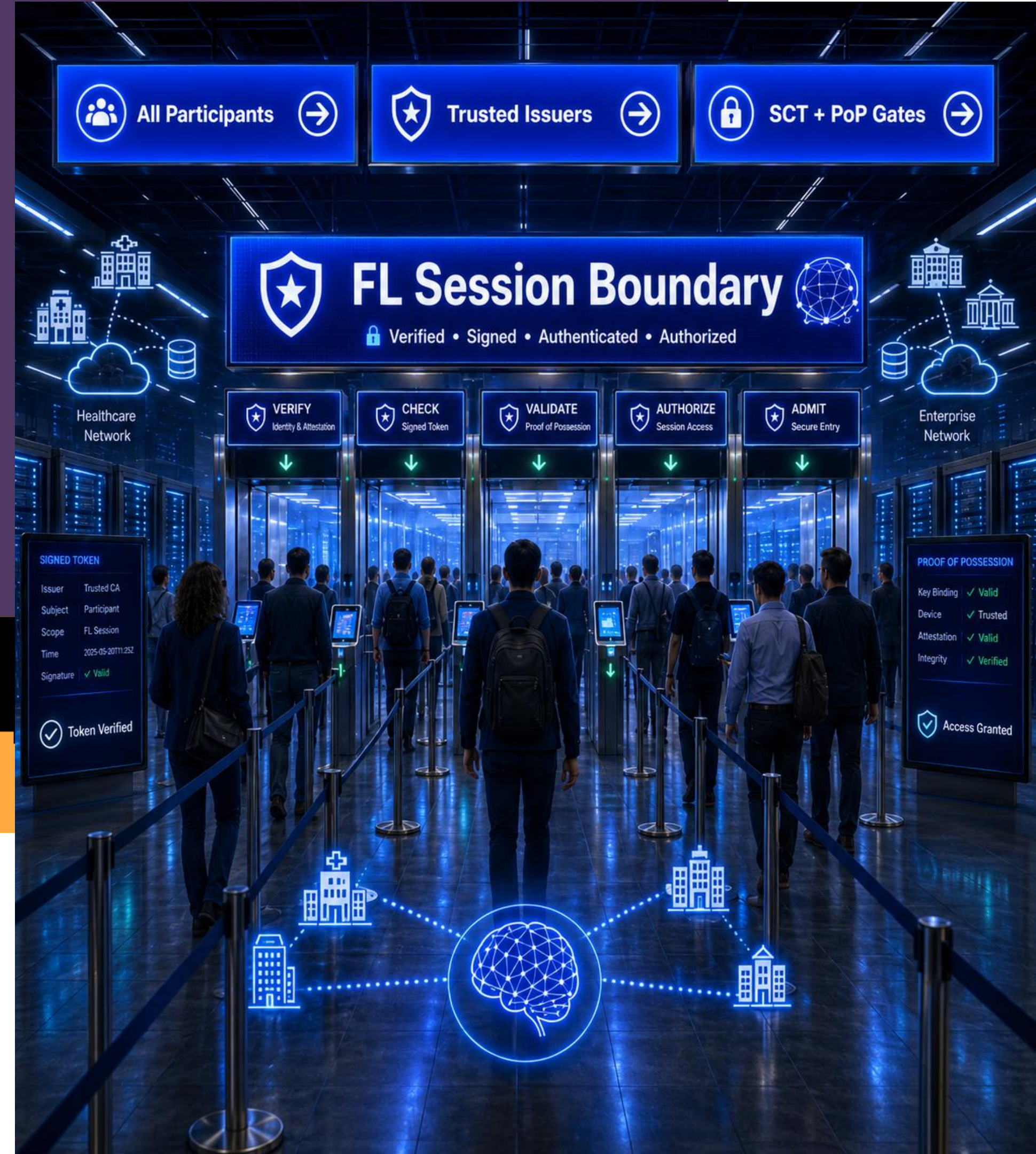
Auditable Session Admission for Cross-Silo Federated Learning

Verifiable entry control before FL orchestration begins



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Session-bound admission control for FL



This talk is about a control point usually assumed rather than modeled:
Admission to the federated session

**Admission
Authentication
Authorization
Auditability**



The problem: FL systems assume federation exists

Cross-silo FL usually starts after admission has already happened.

01

Hidden Assumption

Cross-silo FL usually starts after admission has already happened

02

The Prior Question

Who admitted this participant to this session, under **what** authority, and with **what** proof?

03

Audit Gap

Without explicit admission, later audit can show **what** happened — but not **why** the participant was entitled to enter

- The problem is not **what** happens inside the run, but **who** has the rights to enter it in the first place

Admission Governance is not Operational Governance

- ✓ **Operational controls:** govern what happens inside a session,
- ✓ **Admission controls:** decide who may enter the session boundary

Control Layer	Main Question	Metaphor	Focus
Admission Governance	Who may enter this session boundary?	Passport Access	Entry authority + proof
Operational Governance	What may an admitted participant do inside?	Badge Access	Runtime policy + logs

- Runtime logs can show what happened; admission governance explains why entry was legitimate.

Why existing controls are not enough

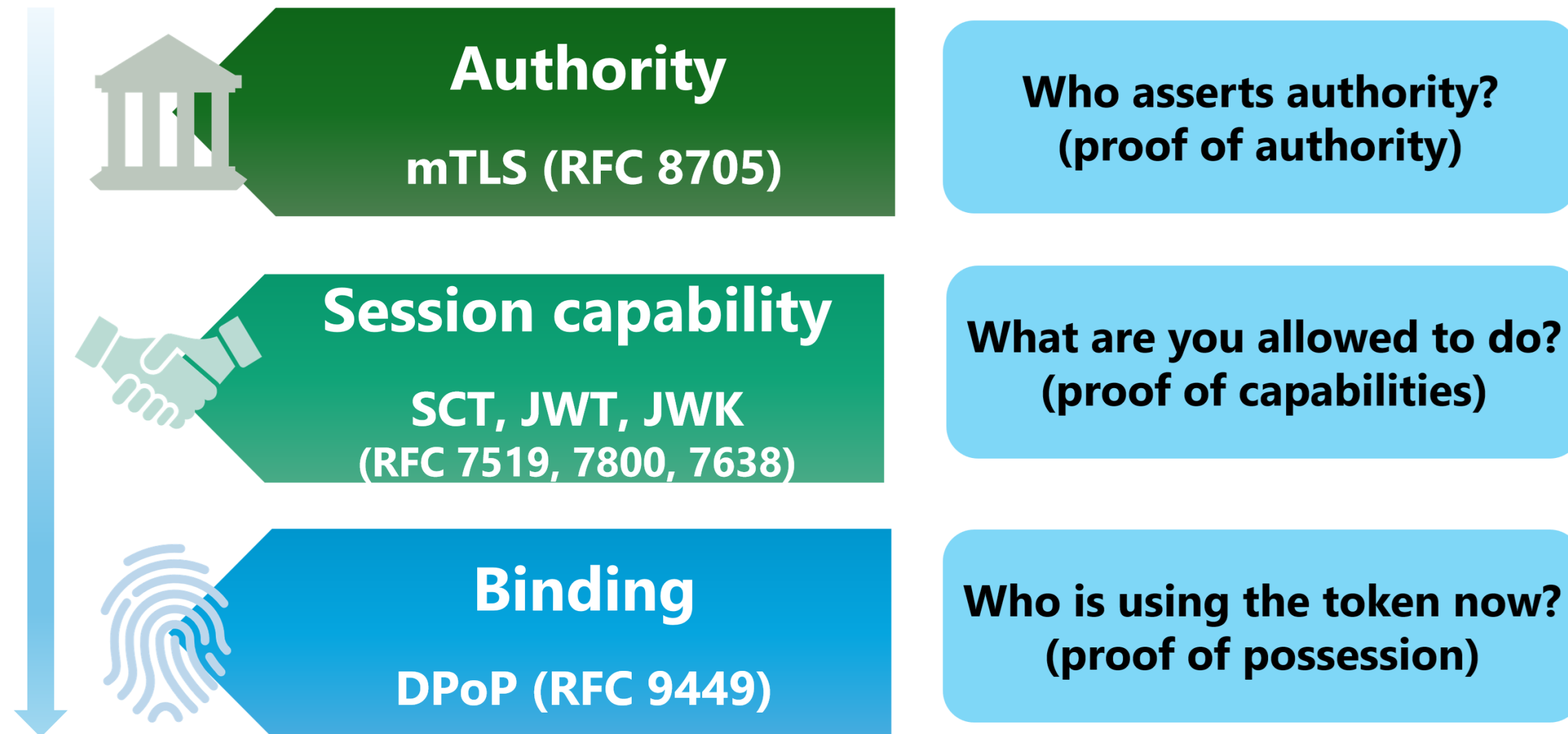
They verify identity, infrastructure, and in-session permissions, **but** do not prove legitimate admission to this specific FL session.

Existing Control	What it answers	What remains unanswered
IAM / authentication	Is this actor known?	Was it admitted to this session?
ABAC / RBAC	What role or attributes does it have	Who granted session-entry authority?
mTLS / Secure channels	Is the connection protected?	Is the participant authorized for this session?
FL orchestration policy	What can happen during training?	Was entry checked before the orchestration began?

- These controls are necessary, but they govern identity, roles, operations, while **admission governance** governs the session-entry boundary.

Solution primitives for proof-carrying session admission

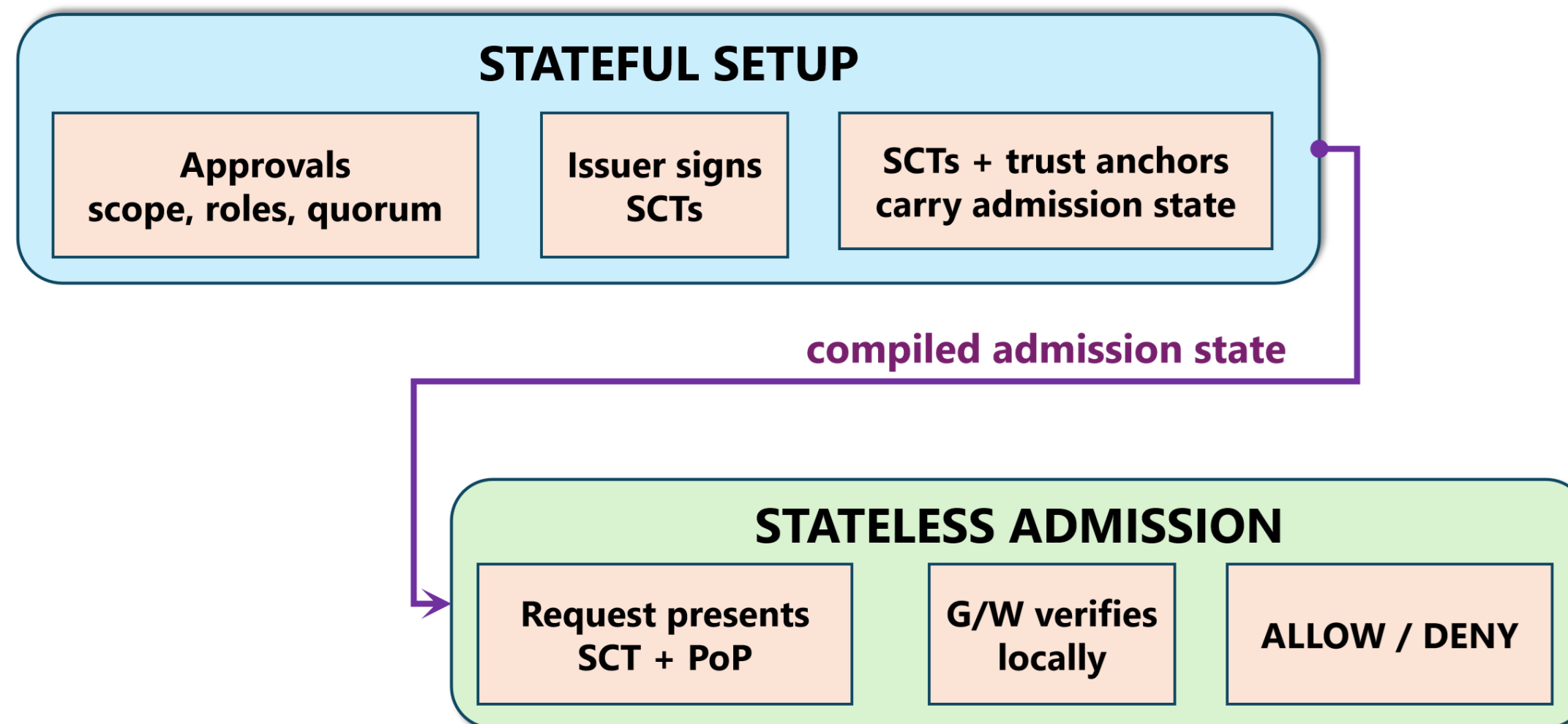
A participant does not merely authenticate; it presents signed authority and proves possession of the bound key.



- The gateway admits a request only when authority, session capability, and possession match.

Solution architecture: stateful setup, stateless admission

- ✓ **Stateful setup** captures approvals, key custody, and token minting
- ✓ **Stateless admission** verifies request-carried proof locally.



- **Portable state**
Signed Session Capability Tokens (SCTs) and trust anchors carry the admission state.
- **Local verification**
The gateway checks SCT + PoP locally, without a central governance lookup.
- **Horizontal scaling**
Gateways can be replicated because they verify rather than negotiate authority.

- State is resolved before the run; runtime admission remains stateless and horizontally scalable.

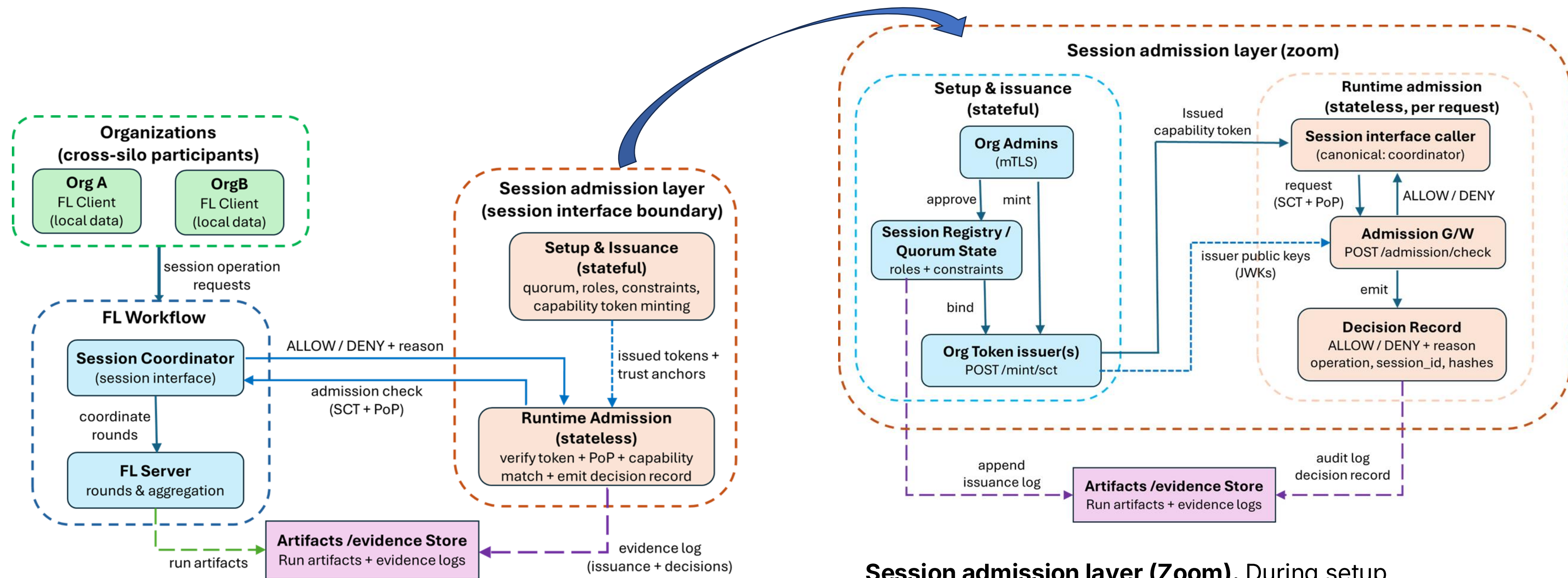
Structured evidence makes admission testable

Each gateway decision becomes an auditable record, not just a runtime side effect.

What is recorded	Why it matters
session_id, decision, reason, timestamp	Shows the admission outcome
request digest + verifier identity	Makes the entry path reconstructible
token/proof validation result	Shows whether authority and possession matched
ALLOW / DENY cases for tampering, mismatch, expiry	Makes the admission model testable

- Our claim is not better model accuracy, but verifiable admission behavior before FL orchestration begins.

<https://github.com/onzelf/FLICS-cross-silo-admission>



Session admission layer (Zoom). During setup, administrators approve quorum roles/constraints and mint SCTs. At runtime, the coordinator submits session requests carrying the SCT and PoP; the admission gateway returns ALLOW/DENY, emits a decision record (operation, session_id, hashes), and appends evidence to the artifact/evidence store.

Admission Governance : The Missing Control Layer



Operations already governed.

FL systems already govern what happens inside the run

Separation of concerns

Distinguishes between "who may enter" (admission) and "what they do inside"

Cryptographic rigor

Replaces implicit trust with a proof-carrying model using SCT and PoP.

Enhanced auditability

Structured evidence ensures every entry decision is a permanent part of the

Orchestration agnostic

The layer protects session entry without replacing FL orchestration.

- Cross-silo FL does not **only** need governance inside the run, it needs admission governance at the session boundary: ***who may enter, under what authority, with what proof, and with what evidence.***



THANK YOU

Any Questions?

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